

DSA 0604

Data Handling and Visualization

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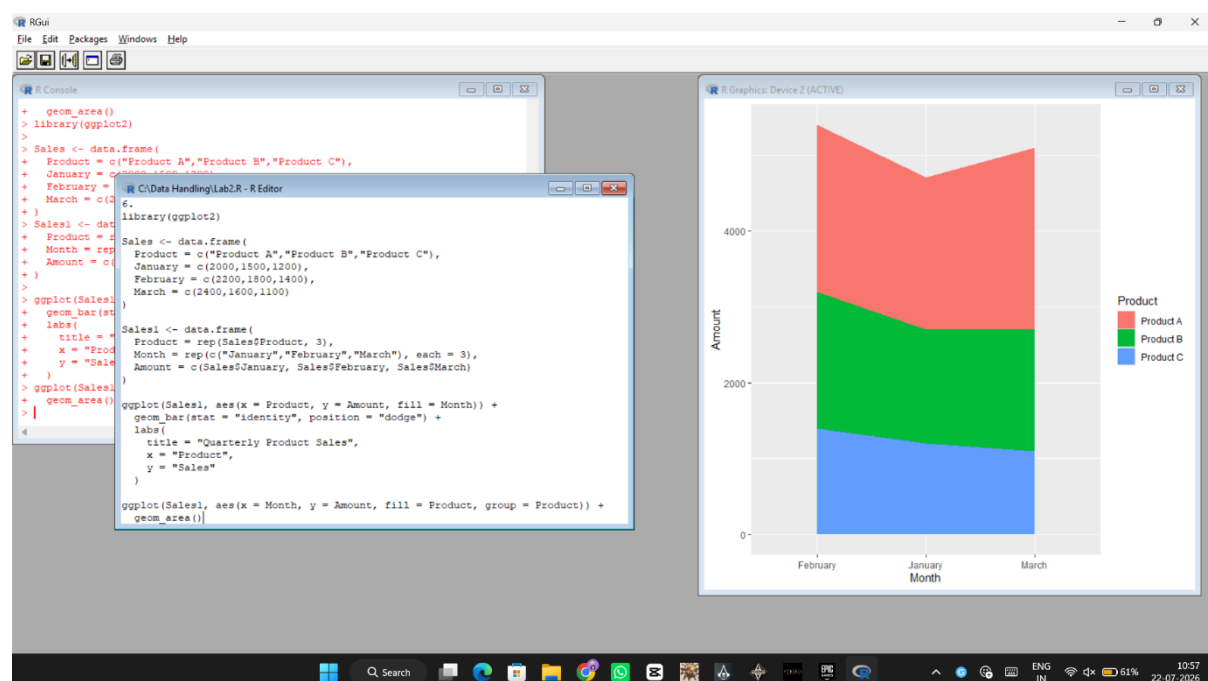
Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID Product Name January Sales February Sales

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.



RGui

File Edit Packages Windows Help

R Console

```

+ Product = rep(Sales$Product, 3),
+ Month = rep(c("January", "February", "March"), each = 3),
+ Amount = c(Sales$January, Sales$February, Sales$March)
+ )
> ggplot(Sales1, aes(x = Product, y = Amount, fill = Month)) +
+   geom_bar(stat = "identity", position = "dodge") +
+   labs(
+     title = "Quarterly Product Sales",
+     x = "Product",
+     y = "Sales"
+   )
> ggplot(Sales1, aes(x = Month, y = Amount, fill = Product, group = Product)) +
+   geom_area()
> Sales
  Product January February March
1 Product A    2000     2200  2400
2 Product B    1500     1800  1600
3 Product C    1200     1400  1100
> Sales
  Product January February March
1 Product A    2000     2200  2400
2 Product B    1500     1800  1600
3 Product C    1200     1400  1100
>

```

C:\Data Handling\Lab2.R - R Editor

```

Sales <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

Sales1 <- data.frame(
  Product = rep(Sales$Product, 3),
  Month = rep(c("January", "February", "March"), each = 3),
  Amount = c(Sales$January, Sales$February, Sales$March)
)

ggplot(Sales1, aes(x = Product, y = Amount, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Quarterly Product Sales",
    x = "Product",
    y = "Sales"
  )

ggplot(Sales1, aes(x = Month, y = Amount, fill = Product, group = Product)) +
  geom_area()

Sales

```

10:58 22-07-2026

RGui

File History Resize Windows

R Console

```

+   geom_area()
> Sales
  Product January Feb
1 Product A    2000
2 Product B    1500
3 Product C    1200
> Sales
  Product January Feb
1 Product A    2000
2 Product B    1500
3 Product C    1200
> Customer <- data.frame(
+   ID = c(1,2,3),
+   Age = c(28,35,42),
+   Gender = c("Female", "Male", "Female"),
+   Income = c(50000, 60000, 75000)
+ )
> ggplot(Customer, aes(x = Age, y = Number of Customers, fill = "skyblue")) +
+   geom_bar()
+   labs(
+     title = "Customer Age Distribution",
+     x = "Age",
+     y = "Number of Customers"
+   )
>

```

C:\Data Handling\Lab2.R - R Editor

```

Sales
library(ggplot2)

Customer <- data.frame(
  ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

ggplot(Customer, aes(x = factor(Age))) +
  geom_bar(fill = "skyblue") +
  labs(
    title = "Customer Age Distribution",
    x = "Age",
    y = "Number of Customers"
  )

```

R Graphics Device 2 (ACTIVE)

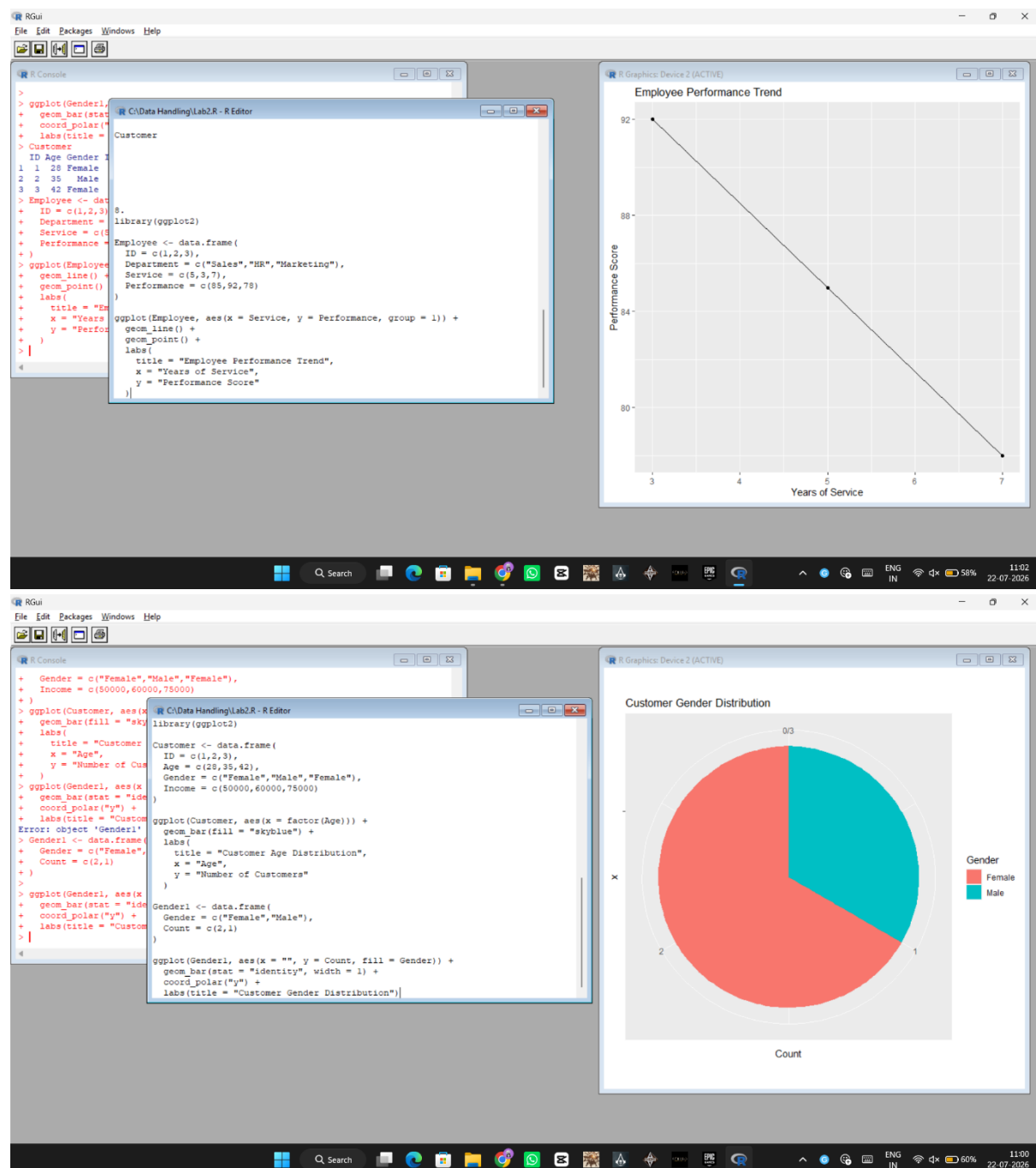
Customer Age Distribution

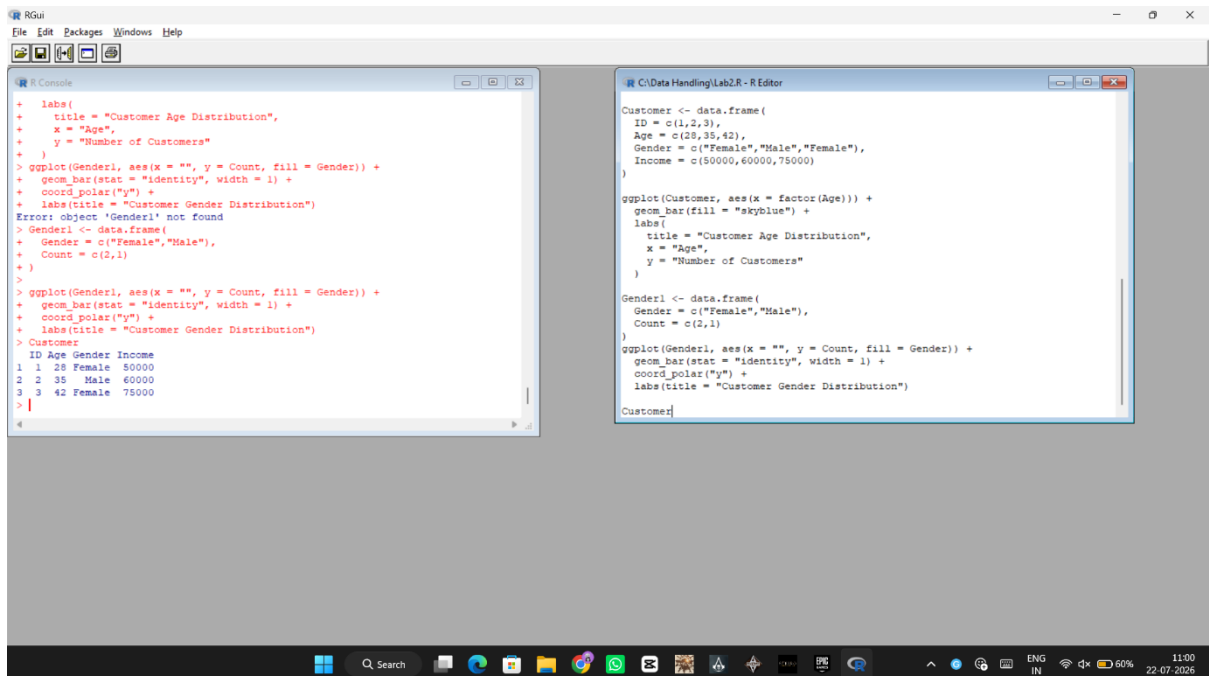
11:00 22-07-2026

Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

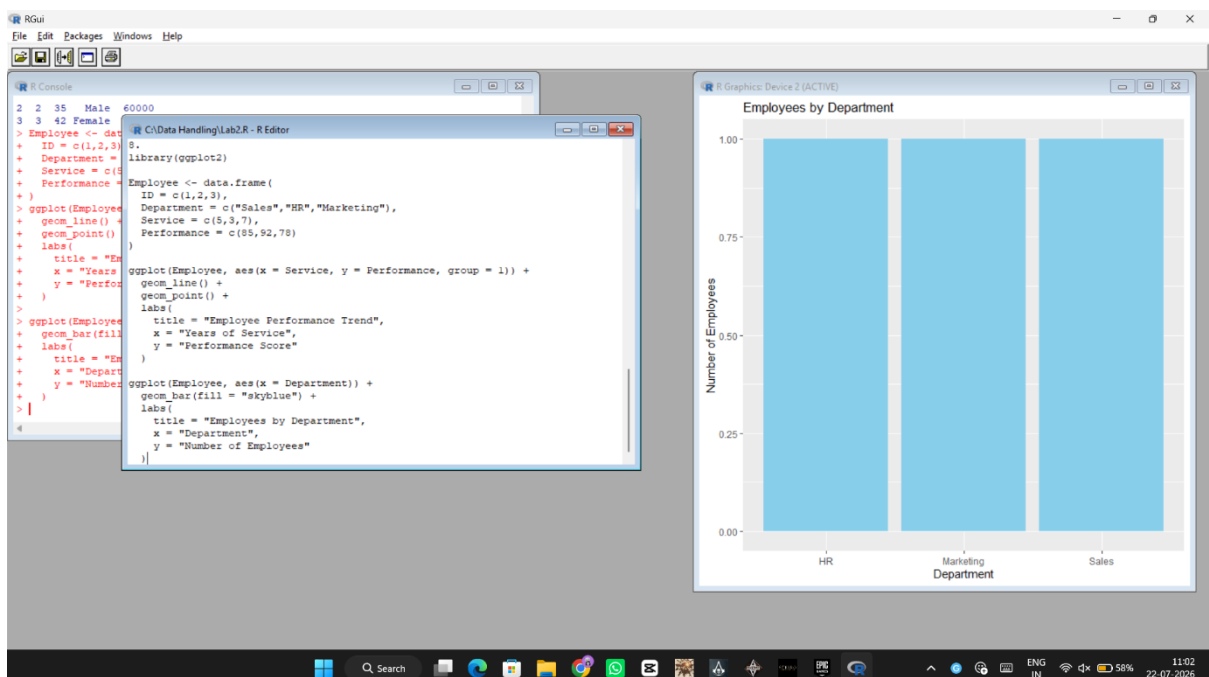
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

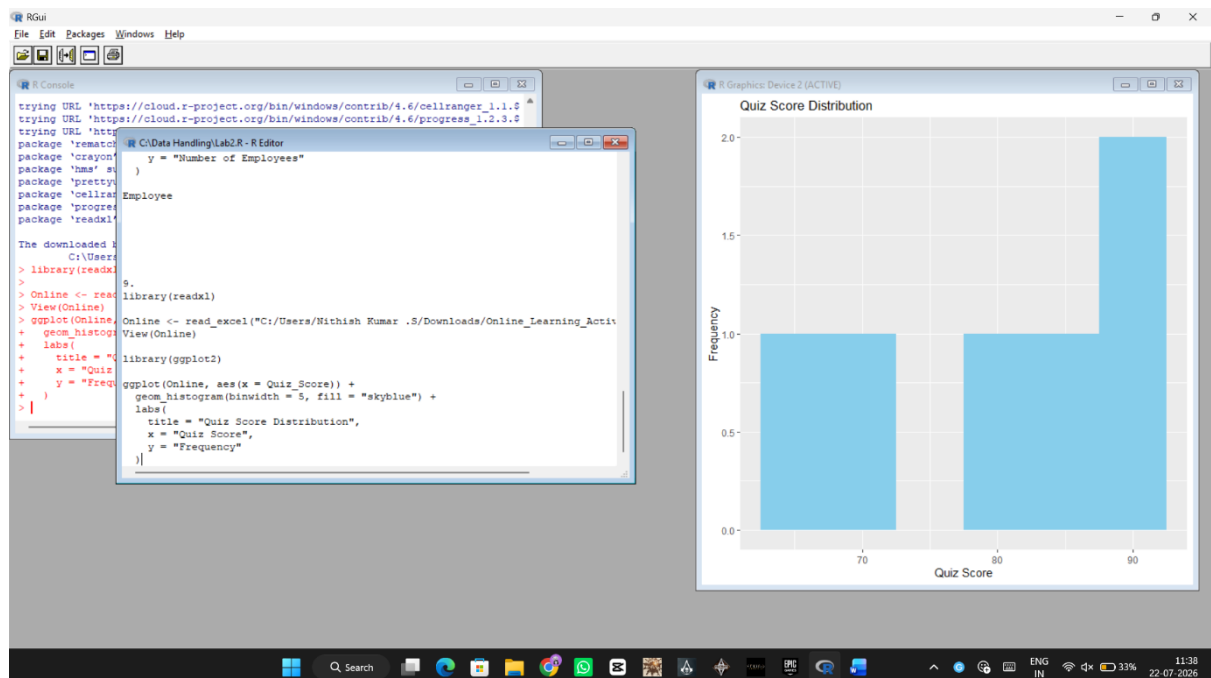
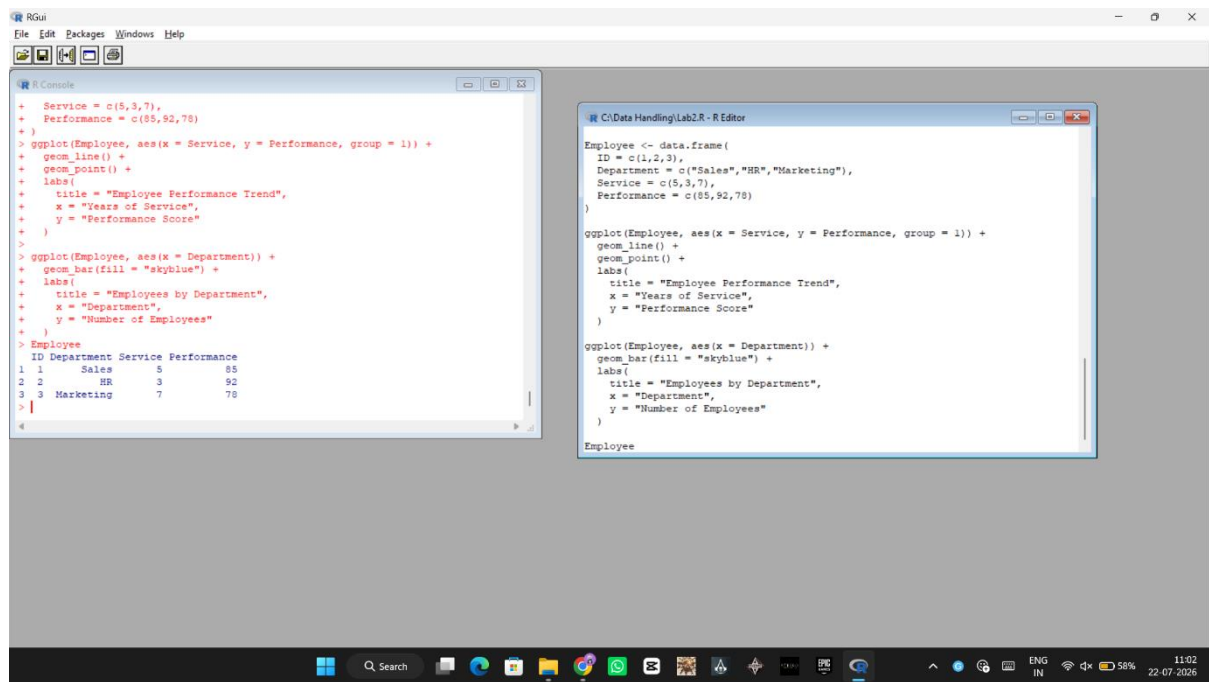




Set 8 Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.





set 9 Dataset: Online_Learning_Activity.csv

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

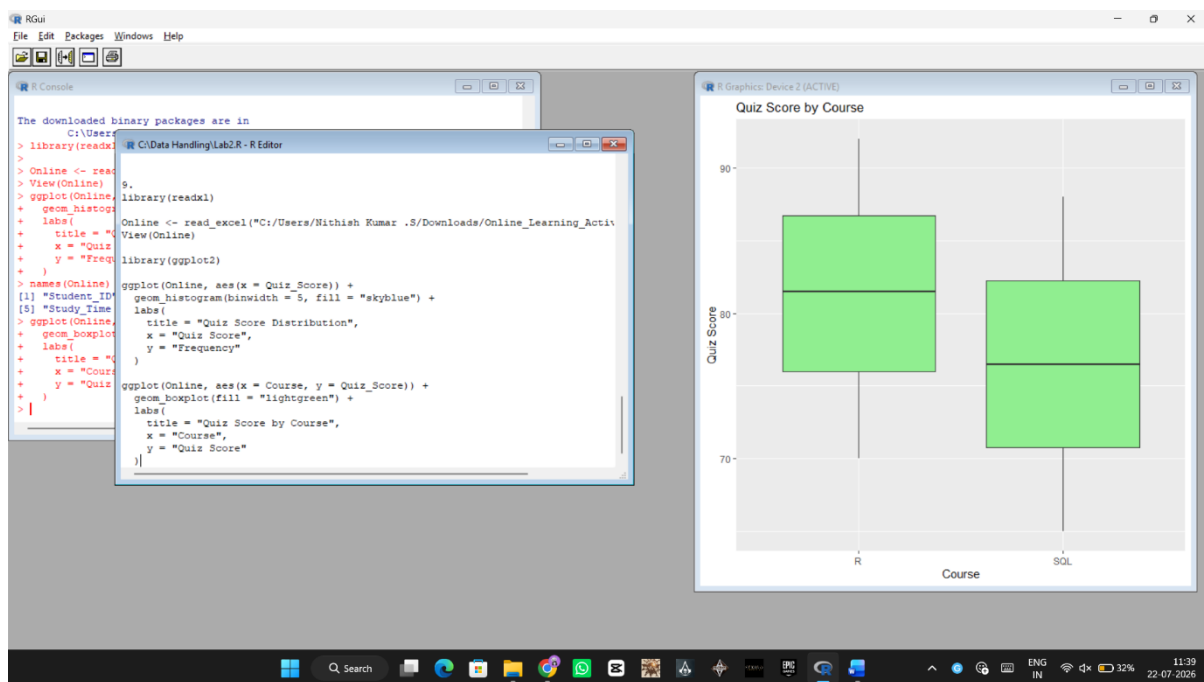
2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent

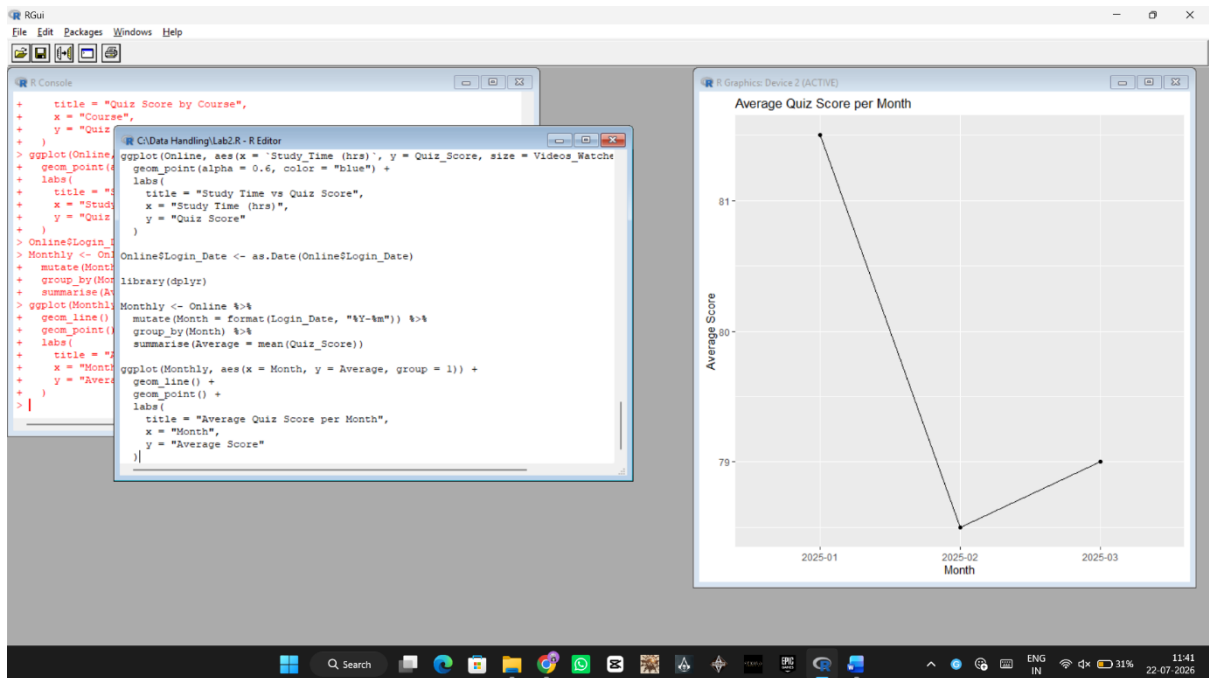
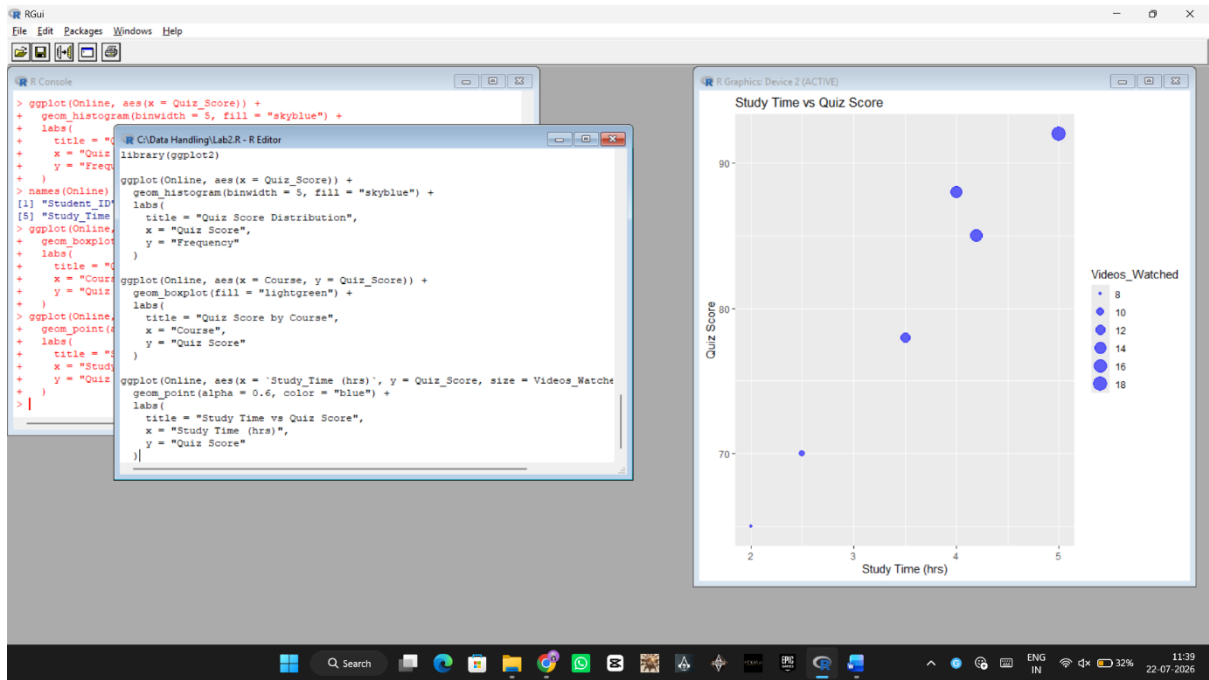
Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per

month. Plot a line chart. Apply moving average smoothing. Identify the trend.

4. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.





Set 10 Survey Responses Analysis

Dataset: Survey Responses

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

